

Stagehand Phase Evidence

Current checkpoint: public Site v88, verified 2026-09-03.

Project

Stagehand is a WebMCP-native 2D animation studio for editable paper-cutout scenes. The current cut supports rigged characters, imported image props, per-asset visual direction, camera work, captions, non-voice audio cues, storyboard beats, multi-scene Preview, and deterministic WebM export.

Private source: github.com/swaymun/stagehand-animation-studio

Commit: 2a8a56cbf823566003a0a58b244ea93cb416bdd4

Public Site: stagehand-animation-studio.saimun-h-shahee.chatgpt.site

Sites version: 88

Acceptance loop

Implement - local tests - Sites deployment - hosted test - UI roast - fixes - second review.

Implemented at this checkpoint

- Revision-safe WebMCP mutations with optional expectedRevision and structured REVISION_CONFLICT results.
- Idempotent WebMCP mutations with optional idempotencyKey replay protection.
- Structured per-asset role, treatment, silhouette, palette, and direction notes with human and agent parity.
- Asset treatment choices reach canvas, thumbnails, Preview, and WebM; new assets also receive valid style defaults immediately.
- Motion actions show affected character and bounded duration; pose presets remain a separate group.
- Human audio cue Start/End editors share bounded timing updates with the agent cue editor.
- Storyboard mode selects the Board rail; Preview hides editing mutation surfaces and selection chrome.

Verification

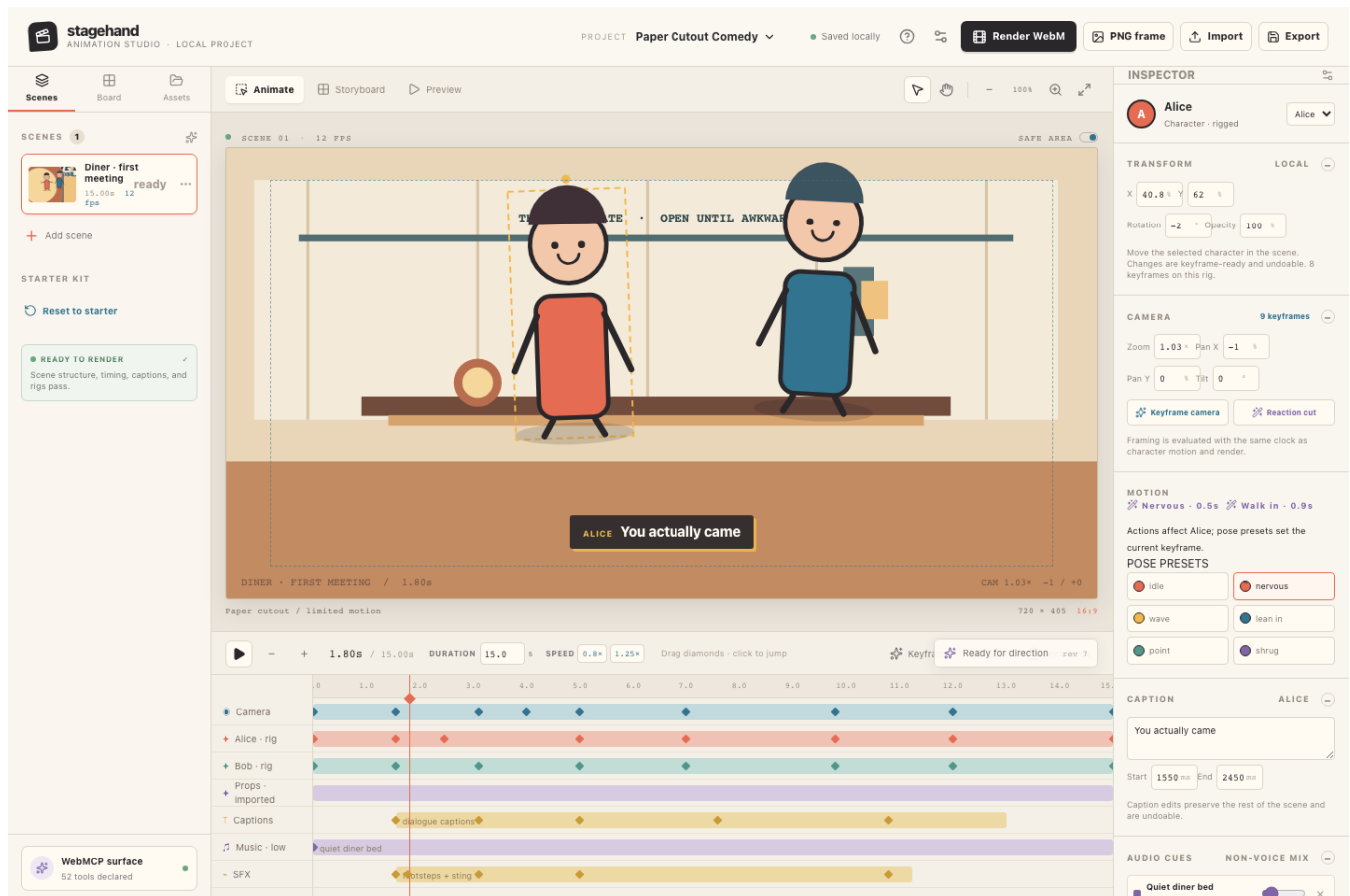
Local gate:

```
npm run format
npm run lint
npm run build
git diff --check
npm run smoke
```

Smoke passed: 52 tools, 39 guarded mutations, a 15-second six-beat starter, synchronized timing, asset treatments, import/recovery including empty collections, Preview, valid PNG/WebM signatures, and zero

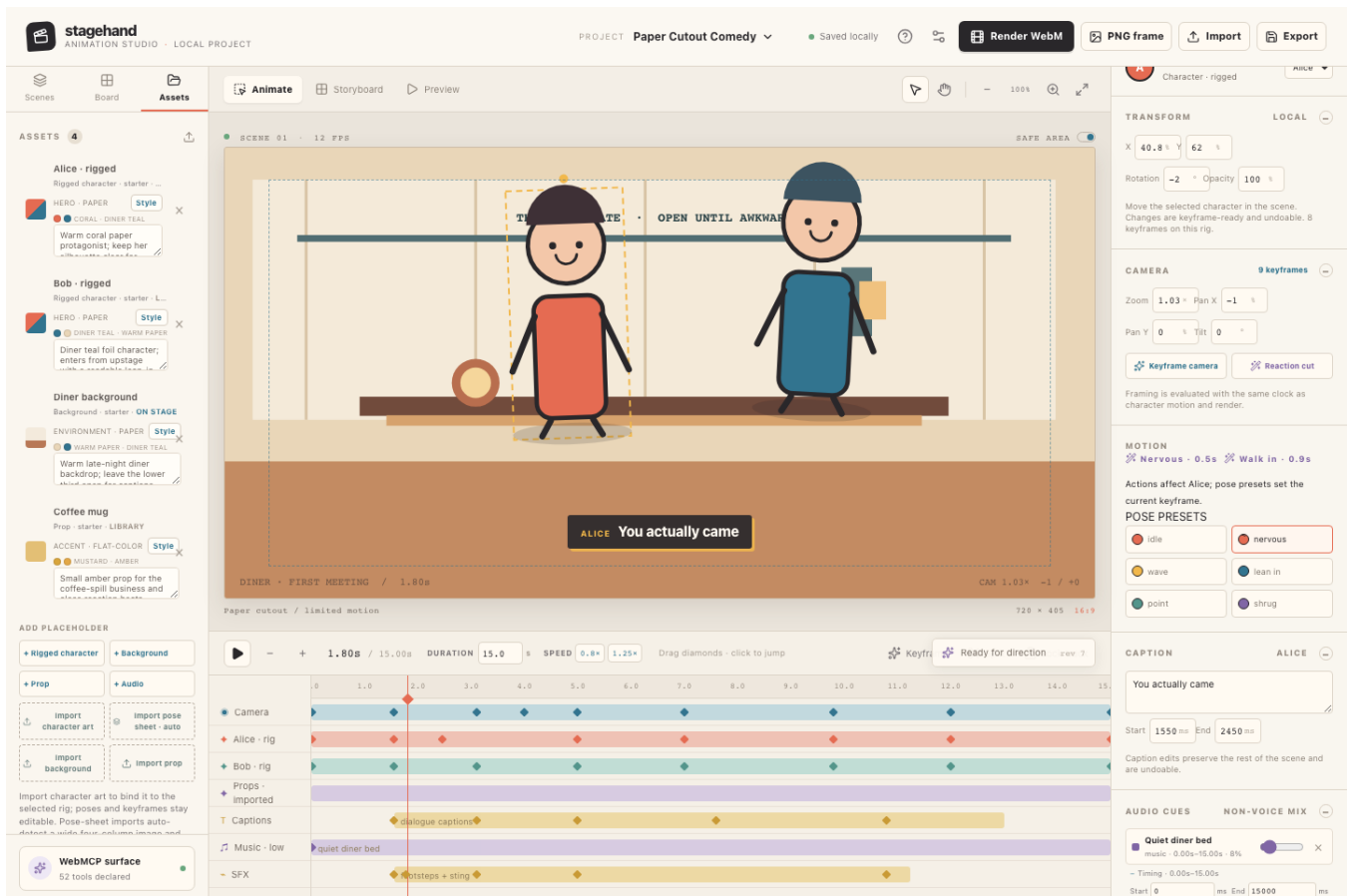
errors.

Animate editing workspace



Hosted Playwright capture at 1440 x 960. The 15-second timeline carries the expanded character and camera beats while speed controls, frame steps, and labels remain readable.

Assets style editor



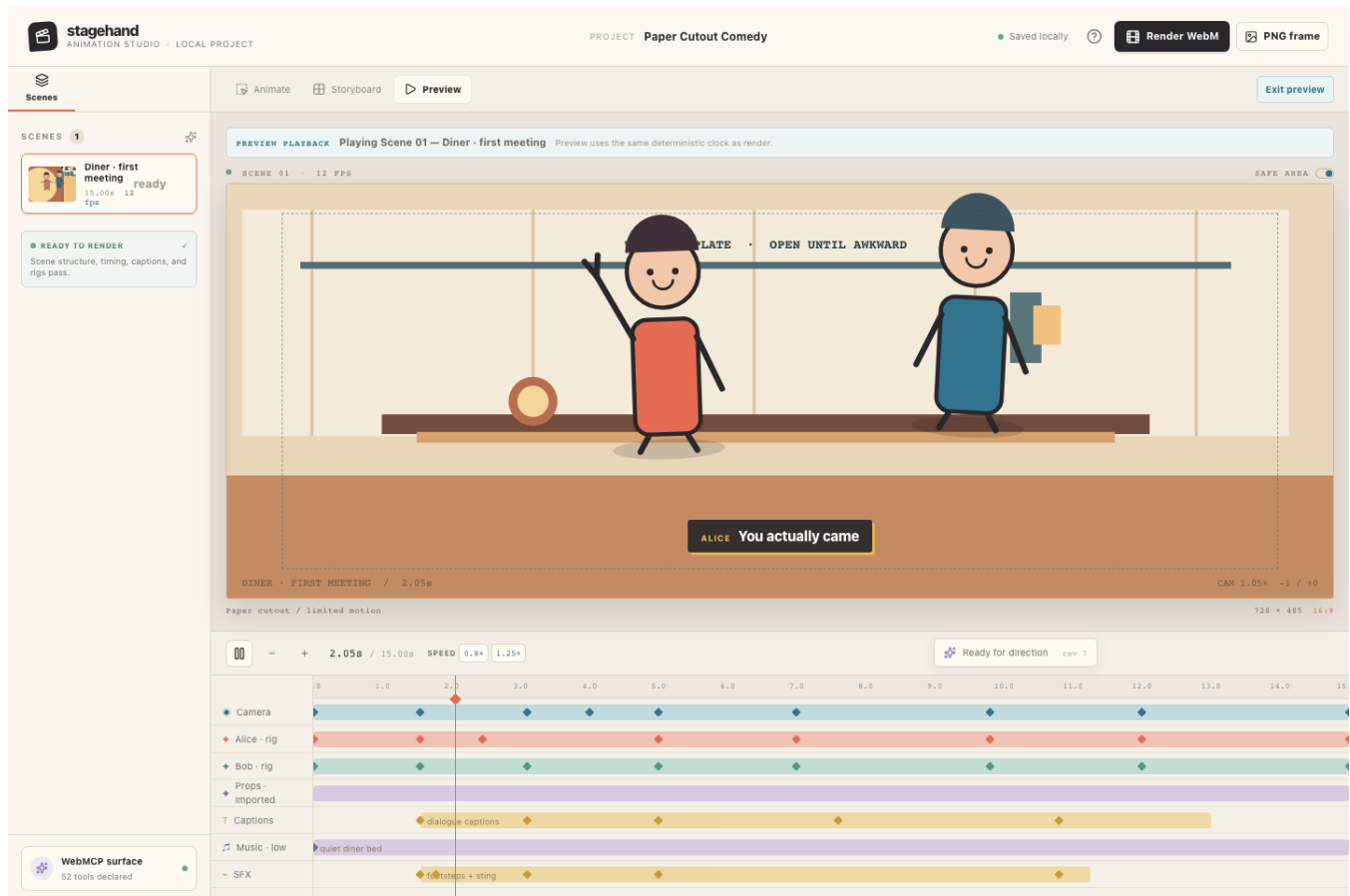
Hosted Playwright capture at 1440 x 960. Palette chips and stage/library cues make the structured Style direction legible without taking over the asset rail.

Storyboard mode with Board rail

The screenshot displays the Stagehand Animation Studio interface in storyboard mode. The top bar shows the project name "Paper Cutout Comedy" and various controls like "Render WebM", "PNG frame", "Import", and "Export". The left sidebar contains a "BEATS" list with six items: "The wait", "The entrance", "The pause", "The spill", "The recovery", and "A second chance". The main area shows a storyboard with six beat cards, each featuring a character illustration and a description. The "The entrance" card is selected, and its details are shown in the right sidebar, including "TRANSFORM" (X: 40.8, Y: 62), "CAMERA" (Zoom: 1.03, Pan X: -1), and "MOTION" (Nervous: 0.5s, Walk in: 0.9s). The bottom of the interface features a "Board rail" with a timeline and a "Keyframe" track, showing the duration of each beat and the timing of keyframes.

Hosted Playwright capture at 1440 x 960. Six beat cards make the longer arc inspectable while the selected mode and project rail stay synchronized.

Review-first Preview player



Hosted Playwright capture at 1440 x 960. Scene-only context, transport, and Exit preview remain; global editing controls are hidden.

Roast findings and fixes

- Imported props lacked direct human transform editing. Added X/Y/Scale/Rot controls sharing the keyframe command path.
- Four prop transform fields initially collided with the asset-list flex rule and truncated. Replaced it with an explicit two-column grid; computed field boxes are 57 px wide in two rows.
- Preview carried too much editing chrome. Converted it into a review-first player with scene-only context and Exit preview.
- Timeline diamonds had 7 x 7 px hit targets. Enlarged the buttons to 22 x 22 pixels while preserving the visual diamond.
- Agent retries could duplicate successful writes. Added optional idempotency-key replay protection and a hosted smoke assertion.
- Switching top-level Storyboard mode could leave Assets selected. Synced mode selection to its companion project rail.
- Character transform inputs lacked explicit accessible names. Added dynamic X/Y/rotation/opacity labels and a smoke check.
- Asset briefs carried style intent only as prose. Added structured role, treatment, silhouette, palette, and direction fields; later made those treatments visible in the canvas and manifest placement cues.
- Motion action buttons were visually mixed with pose presets. Added explicit durations, affected-character copy, and a Pose presets label.
- Agent frame-rate-only render updates reset an existing 1080p choice. Preserved the current resolution when the preset is omitted and added smoke coverage.
- Preview still exposed global project-editing controls. Hid Settings, Import, Export JSON, and rename while retaining WebM and PNG review actions.
- Scene titles competed with the ready badge and were visually shortened. Used a two-line title clamp and positioned the badge outside the text flow.
- The smoke gate did not exercise recovery, modal semantics, or cross-scene playback. Added Help open/close, reload recovery, and sequence Preview transition checks to local and hosted runs.
- The Inspector was a long scroll dump, human scrubbing could leave agent reads one event behind, and keyboard stepping was unreliable after scrub focus. Added native collapsible groups, shared project-view synchronization, a focused stage target, and smoke coverage.
- Export checks only proved that downloads were non-empty. Added binary signature assertions for WebM/EBML and PNG outputs in local and hosted smoke runs.

Limits

Direct CUA inspection was unavailable because the Mac was locked; Playwright was used for hosted interaction and screenshots. The smoke harness proves injected modelContext registration and execution, not live production ChatGPT WebMCP discovery. TTS, voice cloning, dialogue recording, phoneme extraction, and lip-sync remain out of scope.